

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 12 — Nitrogen and Sulfur (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	12.1 Nitrogen Chemistry: Chemical Inertness, Ammonia, Ammonium Salts, Oxides of Nitrogen, Catalytic Cycles
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

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|---|---|
| 12.1 Nitrogen, Ammonia & Nitrogen Oxides | — Lack of reactivity of N ₂ due to high triple bond enthalpy (945 kJ mol ⁻¹) and non-polarity; basicity of ammonia (NH ₃) |
| 12.2 Sulfur Chemistry & Environmental Impact | — Combustion of sulfur-containing fossil fuels forming SO ₂ ; role of SO ₂ and NO ₂ in acid rain formation; environmental impact of sulfur dioxide |
| High-Frequency Core Repeats | — The 10 most frequently examined Cambridge Paper 1 questions on Nitrogen and Sulfur Chemistry (Q101 – Q110). |

SUBTOPIC 12.1 — NITROGEN, AMMONIA & NITROGEN OXIDES (Q1 – Q55)

1 [EASY] Chemical Inertness of Molecular Nitrogen — 9701/11/M/J/23/Q44

[1]

Why is nitrogen gas, N_2 , remarkably unreactive at room temperature and pressure?

- [A] Nitrogen molecules form an impenetrable protective outer electron shell.
- [B] The $N\equiv N$ triple bond has an extraordinarily large bond enthalpy (945 kJ mol^{-1}) and the molecule is completely non-polar, presenting a huge activation energy barrier to reactions.
- [C] Nitrogen has an exceptionally low electronegativity that repels electrophiles.
- [D] Nitrogen atoms possess no non-bonding lone pairs of electrons.

2 [EASY] Formation of Nitrogen Oxides in Car Engines — 9701/12/M/J/23/Q44

[1]

How do oxides of nitrogen (NO and NO_2) form in internal combustion engines of motor vehicles?

- [A] Nitrogen compounds present as chemical additives in the petrol undergo combustion.
- [B] The engine oil reacts with exhaust steam to produce nitrogen gas.
- [C] The catalytic converter oxidises ammonia into nitrogen monoxide.
- [D] The extreme spark temperatures ($> 2000^\circ\text{C}$) provide the activation energy required to break the strong $N\equiv N$ triple bond and allow atmospheric N_2 and O_2 to react directly.

3 [HARD] Homogeneous Catalysis of Acid Rain Formation by NO_2 — 9701/13/M/J/23/Q44

[1]

Nitrogen dioxide, NO_2 , acts as a homogeneous catalyst in the atmospheric oxidation of sulfur dioxide, SO_2 , to sulfur trioxide, SO_3 .

Which pair of equations represents this catalytic cycle?

- [A] Step 1: $SO_2 + NO_2 \rightarrow SO_3 + NO$; Step 2: $2NO + O_2 \rightarrow 2NO_2$
- [B] Step 1: $SO_2 + 2NO_2 \rightarrow SO_4 + 2NO$; Step 2: $2NO + H_2O \rightarrow HNO_2 + HNO_3$
- [C] Step 1: $2SO_2 + NO \rightarrow 2SO_3 + N_2$; Step 2: $N_2 + 2O_2 \rightarrow 2NO_2$
- [D] Step 1: $SO_2 + O_2 \rightarrow SO_4$; Step 2: $SO_4 + NO_2 \rightarrow SO_3 + NO_3$

4 [EASY] Displacement of Ammonia from Ammonium Salts — 9701/11/O/N/23/Q44

[1]

Which reagent and reaction conditions will displace gaseous ammonia from solid ammonium chloride, NH_4Cl ?

- [A] Heating with concentrated sulfuric acid.
- [B] Adding cold dilute hydrochloric acid.
- [C] Warming with solid calcium hydroxide: $2NH_4Cl(s) + Ca(OH)_2(s) \rightarrow CaCl_2(s) + 2H_2O(l) + 2NH_3(g)$
- [D] Passing dry nitrogen gas over the solid.

5 [HARD] Fertilizer Loss: Liming Soil with Ammonium Fertilizers — 9701/12/O/N/23/Q44

[1]

Why must farmers avoid applying slaked lime, $Ca(OH)_2$, to agricultural fields at the same time as ammonium sulfate fertilizer, $(NH_4)_2SO_4$?

- [A] The alkaline calcium hydroxide reacts with ammonium ions to release ammonia gas into the atmosphere, causing loss of essential nitrogen nutrient from the soil.
- [B] Calcium hydroxide forms an insoluble precipitate of calcium ammonium sulfate that suffocates plant roots.
- [C] The mixture catches fire due to extreme heat of neutralisation.
- [D] Ammonium sulfate oxidises calcium into toxic calcium peroxide.

6 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 6 — 9701/11/M/J/22/Q4

[1]

Nitrogen reaction system 6 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $CO(g) + NO(g) \rightarrow C(s) + NO_2(g)$
- [B] $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
- [C] $4NO(g) + CH_4(g) \rightarrow 2N_2(g) + CO_2(g) + 2H_2O(g)$
- [D] $2CO(g) + 2NO(g) \rightarrow 2CO_2(g) + N_2(g)$, converting two harmful pollutants into harmless natural atmospheric gases.

7 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 7 — 9701/12/M/J/222/Q5

[1]

Nitrogen reaction system 7 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

8 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 8 — 9701/13/M/J/223/Q6

[1]

Nitrogen reaction system 8 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

9 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 9 — 9701/11/M/J/224/Q7

[1]

Nitrogen reaction system 9 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

10 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 10 — 9701/12/M/J/220/Q8

[1]

Nitrogen reaction system 10 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

11 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 11 — 9701/13/M/J/221/Q9

[1]

Nitrogen reaction system 11 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

12 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 12 — 9701/11/M/J/222/Q10

[1]

Nitrogen reaction system 12 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

13 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 13 — 9701/12/M/J/223/Q11**[1]**

Nitrogen reaction system 13 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

14 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 14 — 9701/13/M/J/224/Q12**[1]**

Nitrogen reaction system 14 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

15 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 15 — 9701/11/M/J/220/Q13**[1]**

Nitrogen reaction system 15 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

16 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 16 — 9701/12/M/J/221/Q14**[1]**

Nitrogen reaction system 16 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

17 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 17 — 9701/13/M/J/222/Q15**[1]**

Nitrogen reaction system 17 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

18 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 18 — 9701/11/M/J/223/Q16**[1]**

Nitrogen reaction system 18 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

19 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 19 — 9701/12/M/J/224/Q17

[1]

Nitrogen reaction system 19 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

20 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 20 — 9701/13/M/J/220/Q18

[1]

Nitrogen reaction system 20 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

21 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 21 — 9701/11/M/J/221/Q19

[1]

Nitrogen reaction system 21 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

22 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 22 — 9701/12/M/J/222/Q20

[1]

Nitrogen reaction system 22 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

23 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 23 — 9701/13/M/J/223/Q21

[1]

Nitrogen reaction system 23 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

24 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 24 — 9701/11/M/J/224/Q22

[1]

Nitrogen reaction system 24 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

25 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 25 — 9701/12/M/J/220/Q23**[1]**

Nitrogen reaction system 25 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

26 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 26 — 9701/13/M/J/221/Q24**[1]**

Nitrogen reaction system 26 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

27 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 27 — 9701/11/M/J/222/Q25**[1]**

Nitrogen reaction system 27 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

28 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 28 — 9701/12/M/J/223/Q26**[1]**

Nitrogen reaction system 28 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

29 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 29 — 9701/13/M/J/224/Q27**[1]**

Nitrogen reaction system 29 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

30 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 30 — 9701/11/M/J/220/Q28**[1]**

Nitrogen reaction system 30 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

31 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 31 — 9701/12/M/J/221/Q29**[1]**

Nitrogen reaction system 31 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

32 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 32 — 9701/13/M/J/222/Q30**[1]**

Nitrogen reaction system 32 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

33 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 33 — 9701/11/M/J/223/Q31**[1]**

Nitrogen reaction system 33 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

34 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 34 — 9701/12/M/J/224/Q32**[1]**

Nitrogen reaction system 34 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

35 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 35 — 9701/13/M/J/220/Q33**[1]**

Nitrogen reaction system 35 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

36 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 36 — 9701/11/M/J/221/Q34**[1]**

Nitrogen reaction system 36 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

37 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 37 — 9701/12/M/J/222/Q35**[1]**

Nitrogen reaction system 37 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

38 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 38 — 9701/13/M/J/223/Q36**[1]**

Nitrogen reaction system 38 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

39 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 39 — 9701/11/M/J/224/Q37**[1]**

Nitrogen reaction system 39 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

40 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 40 — 9701/12/M/J/220/Q38**[1]**

Nitrogen reaction system 40 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

41 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 41 — 9701/13/M/J/221/Q39**[1]**

Nitrogen reaction system 41 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

42 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 42 — 9701/11/M/J/222/Q40**[1]**

Nitrogen reaction system 42 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum-rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

43 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 43 — 9701/12/M/J/223/Q41**[1]**

Nitrogen reaction system 43 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

44 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 44 — 9701/13/M/J/224/Q42**[1]**

Nitrogen reaction system 44 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

45 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 45 — 9701/11/M/J/220/Q43**[1]**

Nitrogen reaction system 45 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

46 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 46 — 9701/12/M/J/221/Q44**[1]**

Nitrogen reaction system 46 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

47 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 47 — 9701/13/M/J/222/Q45**[1]**

Nitrogen reaction system 47 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

48 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 48 — 9701/11/M/J/223/Q46**[1]**

Nitrogen reaction system 48 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

49 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 49 — 9701/12/M/J/224/Q47**[1]**

Nitrogen reaction system 49 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

50 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 50 — 9701/13/M/J/220/Q48**[1]**

Nitrogen reaction system 50 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

51 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 51 — 9701/11/M/J/221/Q49**[1]**

Nitrogen reaction system 51 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$
- [D] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.

52 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 52 — 9701/12/M/J/222/Q50**[1]**

Nitrogen reaction system 52 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

53 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 53 — 9701/13/M/J/223/Q51**[1]**

Nitrogen reaction system 53 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [B] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

54 [HARD] Nitrogen Chemistry & Atmospheric Dynamics 54 — 9701/11/M/J/224/Q52**[1]**

Nitrogen reaction system 54 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [C] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

55 [EASY] Nitrogen Chemistry & Atmospheric Dynamics 55 — 9701/12/M/J/220/Q53**[1]**

Nitrogen reaction system 55 is evaluated in environmental chemistry. In the automotive catalytic converter, toxic nitrogen monoxide and carbon monoxide are removed simultaneously. What reaction occurs over the platinum–rhodium catalyst?

- [A] $\text{CO(g)} + \text{NO(g)} \rightarrow \text{C(s)} + \text{NO}_2\text{(g)}$
- [B] $2\text{CO(g)} + 2\text{NO(g)} \rightarrow 2\text{CO}_2\text{(g)} + \text{N}_2\text{(g)}$, converting two harmful pollutants into harmless natural atmospheric gases.
- [C] $2\text{NO(g)} + \text{O}_2\text{(g)} \rightarrow 2\text{NO}_2\text{(g)}$
- [D] $4\text{NO(g)} + \text{CH}_4\text{(g)} \rightarrow 2\text{N}_2\text{(g)} + \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$

SUBTOPIC 12.2 — SULFUR CHEMISTRY, ACID RAIN & CONTACT PROCESS (Q56 – Q100)**56 [EASY]** Origin and Environmental Impact of Sulfur Dioxide — 9701/11/M/J/23/Q45**[1]**

What is the primary industrial and fossil–fuel source of atmospheric sulfur dioxide, SO_2 ?

- [A] The reaction of atmospheric nitrogen with sulfuric acid clouds.
- [B] The decomposition of limestone in cement kilns.
- [C] The evaporation of sea spray containing dissolved sulfates.
- [D] The combustion of sulfur-containing fossil fuels (coal and crude oil) in power stations, and the industrial roasting of sulfide metal ores (such as ZnS and FeS_2).

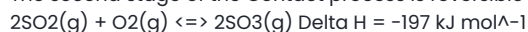
57 [HARD] Flue Gas Desulfurisation (FGD) Chemistry — 9701/12/M/J/23/Q45**[1]**

Coal-fired power stations use flue gas desulfurisation (FGD) systems to prevent sulfur dioxide emissions. What reaction occurs when effluent flue gases are scrubbed with an aqueous slurry of calcium carbonate, CaCO_3 ?

- [A] $\text{CaCO}_3\text{(s)} + \text{SO}_2\text{(g)} \rightarrow \text{CaS(s)} + 2\text{CO}_2\text{(g)} + \text{O}_2\text{(g)}$
- [B] $\text{CaCO}_3\text{(s)} + 2\text{SO}_2\text{(g)} \rightarrow \text{Ca(HSO}_3)_2\text{(aq)} + \text{CO}_2\text{(g)}$
- [C] $\text{CaCO}_3\text{(s)} + \text{SO}_2\text{(g)} \rightarrow \text{CaSO}_3\text{(s)} + \text{CO}_2\text{(g)}$, followed by oxidation to produce calcium sulfate dihydrate (gypsum, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$).
- [D] $\text{CaCO}_3\text{(s)} + \text{SO}_2\text{(g)} + \text{H}_2\text{O(l)} \rightarrow \text{Ca(OH)}_2\text{(s)} + \text{H}_2\text{SO}_4\text{(aq)}$

58 [HARD] Contact Process: Optimum Industrial Compromise Conditions — 9701/13/M/J/23/Q45**[1]**

The second stage of the Contact process is reversible and exothermic:



What are the standard industrial operating conditions and catalyst used for this stage?

- [A] Vanadium(V) oxide (V_2O_5) catalyst, temperature of 450°C , and pressure of 1 – 2 atmospheres.
- [B] Finely divided iron catalyst, temperature of 200°C , and pressure of 200 atmospheres.
- [C] Platinum catalyst, temperature of 800°C , and pressure of 50 atmospheres.
- [D] Nickel catalyst, temperature of 100°C , and pressure of 1 atmosphere.

59 [HARD] Contact Process: Why SO_3 is Not Absorbed Directly in Water — 9701/11/O/N/23/Q45**[1]**

In the final stage of the Contact process, sulfur trioxide, SO_3 , is absorbed into 98% concentrated sulfuric acid rather than directly into pure water.

What is the reason for this industrial procedure?

- [A] Sulfur trioxide does not react with pure water at standard temperatures.
- [B] Direct reaction produces sulfurous acid, H_2SO_3 , instead of sulfuric acid.
- [C] Sulfur trioxide forms an explosive polymer when in contact with pure water.
- [D] Direct reaction of SO_3 with water is violently exothermic, vaporizing the water to produce a dangerous, uncontrollable fog of corrosive sulfuric acid mist that cannot be condensed easily.

60 [EASY] Acid Rain Damage to Limestone Buildings — 9701/12/O/N/23/Q45**[1]**

Acid rain containing sulfuric acid causes severe structural degradation to historical monuments and buildings constructed from limestone or marble (CaCO_3).

Which equation represents this chemical weathering process?

- [A] $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}_2(\text{aq})$
- [B] $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- [C] $\text{CaCO}_3(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Ca}(\text{HSO}_4)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- [D] $2\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{Ca}_2\text{SO}_4(\text{s}) + 2\text{CO}_2(\text{g}) + \text{H}_2(\text{g})$

61 [HARD] Sulfur Chemistry & Industrial Technology 61 — 9701/12/O/N/221/Q36**[1]**

Sulfur reaction cycle 61 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

62 [EASY] Sulfur Chemistry & Industrial Technology 62 — 9701/13/O/N/222/Q37**[1]**

Sulfur reaction cycle 62 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

63 [HARD] Sulfur Chemistry & Industrial Technology 63 — 9701/11/O/N/223/Q38**[1]**

Sulfur reaction cycle 63 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

64 [EASY] Sulfur Chemistry & Industrial Technology 64 — 9701/12/O/N/224/Q39**[1]**

Sulfur reaction cycle 64 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

65 [HARD] Sulfur Chemistry & Industrial Technology 65 — 9701/13/O/N/220/Q40**[1]**

Sulfur reaction cycle 65 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

66 [EASY] Sulfur Chemistry & Industrial Technology 66 — 9701/11/O/N/221/Q41**[1]**

Sulfur reaction cycle 66 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

67 [HARD] Sulfur Chemistry & Industrial Technology 67 — 9701/12/O/N/222/Q42**[1]**

Sulfur reaction cycle 67 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

68 [EASY] Sulfur Chemistry & Industrial Technology 68 — 9701/13/O/N/223/Q43**[1]**

Sulfur reaction cycle 68 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

69 [HARD] Sulfur Chemistry & Industrial Technology 69 — 9701/11/O/N/224/Q44**[1]**

Sulfur reaction cycle 69 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

70 [EASY] Sulfur Chemistry & Industrial Technology 70 — 9701/12/O/N/220/Q45**[1]**

Sulfur reaction cycle 70 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

71 [HARD] Sulfur Chemistry & Industrial Technology 71 — 9701/13/O/N/221/Q46**[1]**

Sulfur reaction cycle 71 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

72 [EASY] Sulfur Chemistry & Industrial Technology 72 — 9701/11/O/N/222/Q47**[1]**

Sulfur reaction cycle 72 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

73 [HARD] Sulfur Chemistry & Industrial Technology 73 — 9701/12/O/N/223/Q48**[1]**

Sulfur reaction cycle 73 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

74 [EASY] Sulfur Chemistry & Industrial Technology 74 — 9701/13/O/N/224/Q49**[1]**

Sulfur reaction cycle 74 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

75 [HARD] Sulfur Chemistry & Industrial Technology 75 — 9701/11/O/N/220/Q50**[1]**

Sulfur reaction cycle 75 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

76 [EASY] Sulfur Chemistry & Industrial Technology 76 — 9701/12/O/N/221/Q51**[1]**

Sulfur reaction cycle 76 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

77 [HARD] Sulfur Chemistry & Industrial Technology 77 — 9701/13/O/N/222/Q52**[1]**

Sulfur reaction cycle 77 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

78 [EASY] Sulfur Chemistry & Industrial Technology 78 — 9701/11/O/N/223/Q53**[1]**

Sulfur reaction cycle 78 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

79 [HARD] Sulfur Chemistry & Industrial Technology 79 — 9701/12/O/N/224/Q54**[1]**

Sulfur reaction cycle 79 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

80 [EASY] Sulfur Chemistry & Industrial Technology 80 — 9701/13/O/N/220/Q55**[1]**

Sulfur reaction cycle 80 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

81 [HARD] Sulfur Chemistry & Industrial Technology 81 — 9701/11/O/N/221/Q56**[1]**

Sulfur reaction cycle 81 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

82 [EASY] Sulfur Chemistry & Industrial Technology 82 — 9701/12/O/N/222/Q57**[1]**

Sulfur reaction cycle 82 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

83 [HARD] Sulfur Chemistry & Industrial Technology 83 — 9701/13/O/N/223/Q58**[1]**

Sulfur reaction cycle 83 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

84 [EASY] Sulfur Chemistry & Industrial Technology 84 — 9701/11/O/N/224/Q59**[1]**

Sulfur reaction cycle 84 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

85 [HARD] Sulfur Chemistry & Industrial Technology 85 — 9701/12/O/N/220/Q60**[1]**

Sulfur reaction cycle 85 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

86 [EASY] Sulfur Chemistry & Industrial Technology 86 — 9701/13/O/N/221/Q61**[1]**

Sulfur reaction cycle 86 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

87 [HARD] Sulfur Chemistry & Industrial Technology 87 — 9701/11/O/N/222/Q62**[1]**

Sulfur reaction cycle 87 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

88 [EASY] Sulfur Chemistry & Industrial Technology 88 — 9701/12/O/N/223/Q63**[1]**

Sulfur reaction cycle 88 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

89 [HARD] Sulfur Chemistry & Industrial Technology 89 — 9701/13/O/N/224/Q64**[1]**

Sulfur reaction cycle 89 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

90 [EASY] Sulfur Chemistry & Industrial Technology 90 — 9701/11/O/N/220/Q65**[1]**

Sulfur reaction cycle 90 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

91 [HARD] Sulfur Chemistry & Industrial Technology 91 — 9701/12/O/N/221/Q66**[1]**

Sulfur reaction cycle 91 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)

92 [EASY] Sulfur Chemistry & Industrial Technology 92 — 9701/13/O/N/222/Q67**[1]**

Sulfur reaction cycle 92 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

93 [HARD] Sulfur Chemistry & Industrial Technology 93 — 9701/11/O/N/223/Q68**[1]**

Sulfur reaction cycle 93 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [B] H_2SO_5 (peroxomonosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

94 [EASY] Sulfur Chemistry & Industrial Technology 94 — 9701/12/O/N/224/Q69**[1]**

Sulfur reaction cycle 94 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

95 [HARD] Sulfur Chemistry & Industrial Technology 95 — 9701/13/O/N/220/Q70**[1]**

Sulfur reaction cycle 95 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO_3 into concentrated H_2SO_4 ?

- [A] H_2SO_5 (peroxomonosulfuric acid)
- [B] $\text{H}_2\text{S}_2\text{O}_7$ (disulfuric acid)
- [C] $\text{H}_2\text{S}_2\text{O}_3$ (thiosulfuric acid)
- [D] $\text{H}_2\text{S}_2\text{O}_8$ (peroxodisulfuric acid)

96 [EASY] Sulfur Chemistry & Industrial Technology 96 — 9701/11/O/N/221/Q71**[1]**

Sulfur reaction cycle 96 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO₃ into concentrated H₂SO₄?

- [A] H₂SO₅ (peroxomonosulfuric acid)
- [B] H₂S₂O₃ (thiosulfuric acid)
- [C] H₂S₂O₈ (peroxodisulfuric acid)
- [D] H₂S₂O₇ (disulfuric acid)

97 [HARD] Sulfur Chemistry & Industrial Technology 97 — 9701/12/O/N/222/Q72**[1]**

Sulfur reaction cycle 97 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO₃ into concentrated H₂SO₄?

- [A] H₂SO₅ (peroxomonosulfuric acid)
- [B] H₂S₂O₃ (thiosulfuric acid)
- [C] H₂S₂O₇ (disulfuric acid)
- [D] H₂S₂O₈ (peroxodisulfuric acid)

98 [EASY] Sulfur Chemistry & Industrial Technology 98 — 9701/13/O/N/223/Q73**[1]**

Sulfur reaction cycle 98 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO₃ into concentrated H₂SO₄?

- [A] H₂S₂O₇ (disulfuric acid)
- [B] H₂SO₅ (peroxomonosulfuric acid)
- [C] H₂S₂O₃ (thiosulfuric acid)
- [D] H₂S₂O₈ (peroxodisulfuric acid)

99 [HARD] Sulfur Chemistry & Industrial Technology 99 — 9701/11/O/N/224/Q74**[1]**

Sulfur reaction cycle 99 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO₃ into concentrated H₂SO₄?

- [A] H₂SO₅ (peroxomonosulfuric acid)
- [B] H₂S₂O₃ (thiosulfuric acid)
- [C] H₂S₂O₈ (peroxodisulfuric acid)
- [D] H₂S₂O₇ (disulfuric acid)

100 [EASY] Sulfur Chemistry & Industrial Technology 100 — 9701/12/O/N/220/Q75**[1]**

Sulfur reaction cycle 100 is monitored. What is the chemical formula of oleum (fuming sulfuric acid), produced by absorbing SO₃ into concentrated H₂SO₄?

- [A] H₂SO₅ (peroxomonosulfuric acid)
- [B] H₂S₂O₇ (disulfuric acid)
- [C] H₂S₂O₃ (thiosulfuric acid)
- [D] H₂S₂O₈ (peroxodisulfuric acid)

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)**101** [EASY] [HIGH-FREQUENCY EXAM REPEAT] Bond Enthalpy and Inertness of Molecular Nitrogen — 9701/11/M/J/23/Q45**[1]**

Why is nitrogen gas, N₂, chemically unreactive under normal atmospheric conditions?

- [A] Nitrogen molecules form intermolecular hydrogen bonds in the gas phase.
- [B] Nitrogen atoms have an expanded octet of 10 valence electrons.
- [C] The N#N triple bond has a very high bond dissociation enthalpy (945 kJ mol⁻¹), and the lack of bond polarity prevents electrophilic and nucleophilic attack.
- [D] Nitrogen gas is denser than air, preventing molecular collisions.

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Catalytic Role of NO₂ in Atmospheric Acid Rain Formation — 9701/12/M/J/23/Q45**[1]**

How does nitrogen dioxide, NO₂, catalyse the oxidation of sulfur dioxide, SO₂, in the atmosphere?

- [A] NO₂ oxidises SO₂ to SO₃ while being reduced to NO; the NO is then re-oxidised back to NO₂ by atmospheric oxygen, regenerating the catalyst.
- [B] NO₂ dissolves in water to form nitric acid, which acts as a Brønsted-Lowry catalyst.
- [C] NO₂ acts as a heterogeneous surface adsorbent for sulfur molecules.
- [D] NO₂ absorbs ultraviolet light to emit high-energy gamma radiation.

103 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Displacement of Ammonia from Ammonium Salts — 9701/13/M/J/23/Q45**[1]**

What observation confirms the evolution of ammonia gas when an ammonium salt is heated with aqueous sodium hydroxide?

- [A] A brown gas is evolved that relights a glowing splint.
- [B] A colorless odorless gas is evolved that turns limewater cloudy.
- [C] A dense white precipitate is formed in the gas phase.
- [D] A pungent choking gas is evolved that turns damp red litmus paper blue.

104 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Formation of Nitrogen Monoxide in Car Engines — 9701/11/O/N/23/Q46**[1]**

Why is nitrogen monoxide, NO, formed in car internal combustion engines but not under ambient room temperature conditions?

- [A] Petrol contains liquid nitrogen monoxide as a fuel additive.
- [B] The reaction $\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}(\text{g})$ is highly endothermic and has a massive activation energy that can only be overcome at the intense temperatures ($> 2000\text{ }^\circ\text{C}$) inside engine cylinders.
- [C] Air inside car engines is under reduced pressure, favoring NO formation.
- [D] The spark plug acts as a heterogeneous catalyst for nitrogen dissociation.

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Flue Gas Desulfurisation Chemistry — 9701/12/O/N/23/Q46**[1]**

In a coal-fired power station, flue gas desulfurisation uses calcium carbonate to remove sulfur dioxide. What is the commercial product formed after aeration of the resulting sludge?

- [A] Gypsum, CaSO₄·2H₂O (calcium sulfate dihydrate), used in construction plasterboard.
- [B] Quicklime, CaO, used for road paving.
- [C] Slaked lime, Ca(OH)₂, used for water softening.
- [D] Calcium carbide, CaC₂, used to produce acetylene gas.

106 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Industrial Reason for Oleum Formation in the Contact Process — 9701/13/O/N/23/Q46**[1]**

In the manufacture of sulfuric acid by the Contact process, why is sulfur trioxide absorbed in 98% sulfuric acid rather than water?

- [A] Water decomposes sulfur trioxide into sulfur dioxide and oxygen.
- [B] The reaction with water has an activation energy that requires temperatures above 800 °C.
- [C] The direct reaction with water is so exothermic that it vaporizes the water, producing a hazardous, uncontrollable fog of sulfuric acid droplets that is difficult to condense.
- [D] Sulfuric acid acts as a homogeneous catalyst for water dissociation.

107 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Chemical Weathering of Marble Statues by Acid Rain — 9701/11/F/M/24/Q34**[1]**

Which chemical equation represents the reaction of acid rain containing sulfuric acid with marble statues (CaCO₃)?

- [A] $\text{CaCO}_3(\text{s}) + \text{SO}_2(\text{aq}) \rightarrow \text{CaSO}_3(\text{s}) + \text{CO}_2(\text{g})$
- [B] $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- [C] $\text{CaCO}_3(\text{s}) + 2\text{HNO}_3(\text{aq}) \rightarrow \text{Ca}(\text{NO}_3)_2(\text{aq}) + \text{H}_2(\text{g}) + \text{CO}_2(\text{g})$
- [D] $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaS}(\text{s}) + 2\text{O}_2(\text{g}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Automotive Catalytic Converter Reductions and Oxidations — 9701/12/F/M/24/Q34**[1]**

In a modern three-way vehicle catalytic converter, which two simultaneous conversions take place?

- [A] Reduction of CO to carbon; Oxidation of NO to NO₂.
- [B] Oxidation of SO₂ to SO₃; Reduction of CO₂ to methane.
- [C] Reduction of H₂O to hydrogen gas; Oxidation of N₂ to nitrates.
- [D] Oxidation of CO and unburnt hydrocarbons to CO₂ and H₂O; Reduction of NO to N₂.

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Chemical Formula and Dilution of Oleum — 9701/13/F/M/24/Q34**[1]**

What is the chemical formula of oleum, and what product is formed when oleum is mixed with water in a 1 : 1 mole ratio?

- [A] Oleum is H₂S₂O₇; mixing 1 mole of H₂S₂O₇ with 1 mole of H₂O produces 2 moles of pure H₂SO₄.
- [B] Oleum is H₂SO₅; mixing with water produces H₂SO₄ and H₂O₂.
- [C] Oleum is H₂S₂O₈; mixing with water produces H₂SO₄ and O₂.
- [D] Oleum is H₂S₂O₃; mixing with water produces sulfur and SO₂.

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Bond Angle Change During Ammonia Protonation — 9701/11/M/J/22/Q38**[1]**

What is the H-N-H bond angle in ammonia, NH₃, and what happens to this bond angle when NH₃ reacts with a proton (H⁺) to form the ammonium ion, NH₄⁺?

- [A] NH₃ has a bond angle of 109.5°; upon protonation, the angle decreases to 104.5°.
- [B] NH₃ has a bond angle of 120.0°; upon protonation, the angle becomes 180.0°.
- [C] NH₃ has a bond angle of 107.0° (trigonal pyramidal); upon protonation, the angle increases to 109.5° (regular tetrahedral).
- [D] NH₃ has a bond angle of 107.0°; upon protonation, the angle decreases to 90.0°.

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. The N ₂ molecule contains a triple covalent bond (one sigma and two pi bonds) that requires 945 kJ/mol to break. Furthermore, because both atoms are identical, the molecule is completely non-polar with no permanent dipole to attract polar attacking reagents or electrophiles/nucleophiles. This combination imparts high chemical inertia under normal conditions.	[1]
2	D	Option D is correct. Petrol itself contains negligible nitrogen. However, the high-voltage electrical spark and high compression temperatures inside car engine cylinders exceed 2000 °C, supplying sufficient kinetic activation energy to overcome the 945 kJ/mol barrier of N ₂ : N ₂ (g) + O ₂ (g) → 2NO(g). Once released into the atmosphere, NO cools and oxidises further: 2NO + O ₂ → 2NO ₂ .	[1]
3	A	Option A is correct. Gaseous SO ₂ reacts very slowly with oxygen directly. Nitrogen dioxide acts as a homogeneous atmospheric catalyst: Step 1: SO ₂ (g) + NO ₂ (g) → SO ₃ (g) + NO(g) (NO ₂ oxidises SO ₂ to SO ₃ and is reduced to NO). Step 2: 2NO(g) + O ₂ (g) → 2NO ₂ (g) (NO is rapidly re-oxidised by atmospheric oxygen to regenerate the NO ₂ catalyst). The produced SO ₃ dissolves in rain to form H ₂ SO ₄ .	[1]
4	C	Option C is correct. Ammonia is a weak base, so any strong base (such as Ca(OH) ₂ or NaOH) will displace ammonia from its salts when heated: NH ₄ ⁺ + OH ⁻ → NH ₃ (g) + H ₂ O. Gaseous ammonia is evolved (recognized by its pungent smell and turning damp red litmus paper blue).	[1]
5	A	Option A is correct. Adding basic calcium hydroxide (slaked lime) to soil raises the pH (high [OH ⁻]). If ammonium fertilizer is present, an acid-base displacement occurs: (NH ₄) ₂ SO ₄ + Ca(OH) ₂ → CaSO ₄ + 2H ₂ O + 2NH ₃ (g). The nitrogen is lost as volatile gaseous ammonia into the air, wasting expensive fertilizer and causing atmospheric pollution.	[1]
6	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
7	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
8	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
9	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
10	A	Option A is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
11	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
12	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
13	A	Option A is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
14	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
15	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
16	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
17	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
18	A	Option A is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
19	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
20	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
21	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
22	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]
23	A	Option A is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: 2CO + 2NO → 2CO ₂ + N ₂ .	[1]

23

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
51	D	Option D is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
52	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
53	A	Option A is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
54	C	Option C is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
55	B	Option B is correct. Modern catalytic converters eliminate carbon monoxide (toxic asphyxiant) and nitrogen monoxide (pollutant and acid rain precursor) simultaneously via the heterogeneous catalysed reaction: $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$.	[1]
56	D	Option D is correct. Fossil fuels (especially coal and heavy crude oil) contain natural organosulfur and inorganic iron sulfide impurities. When burned, sulfur is oxidized to sulfur dioxide: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$. Additionally, metallurgical extraction of metals like zinc and lead involves roasting sulfide ores in air: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$.	[1]
57	C	Option C is correct. In FGD scrubbers, acidic SO_2 reacts with basic calcium carbonate slurry to form insoluble calcium sulfite: $\text{CaCO}_3 + \text{SO}_2 \rightarrow \text{CaSO}_3 + \text{CO}_2$. Air is blown through the slurry to oxidise CaSO_3 into calcium sulfate: $2\text{CaSO}_3 + \text{O}_2 + 4\text{H}_2\text{O} \rightarrow 2(\text{CaSO}_4 \cdot 2\text{H}_2\text{O})$. This byproduct is synthetic gypsum, widely sold for manufacturing plasterboard and cement.	[1]
58	A	Option A is correct. The forward reaction is exothermic and produces fewer moles of gas ($3 \rightarrow 2$). Low temperature favors high yield, but would give an unacceptably slow rate; 450°C is the compromise temperature. The catalyst is vanadium(V) oxide, V_2O_5 . Even at 1–2 atm, conversion is already ~98%, so expensive high-pressure plant equipment is unnecessary.	[1]
59	D	Option D is correct. The direct reaction $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$ is intensely exothermic. The enormous heat release immediately boils the water, creating a persistent, hazardous aerosol mist of tiny sulfuric acid droplets that pass right through condenser pipes into the atmosphere. Instead, SO_3 is smoothly absorbed into 98% H_2SO_4 to form liquid oleum: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. The oleum is then safely diluted with calculated water: $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$.	[1]
60	B	Option B is correct. Sulfuric acid in acid rain reacts directly with calcium carbonate in limestone/marble: $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$. The calcium sulfate formed is more soluble than calcium carbonate and crumbles away, eroding fine architectural details.	[1]
61	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
62	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
63	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
64	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
65	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
66	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
67	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
68	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
69	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
70	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
71	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
72	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
73	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
74	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
75	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
76	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
77	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
78	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
79	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
80	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
81	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
82	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
83	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
84	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
85	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
86	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
87	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
88	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
89	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
90	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
91	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
92	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
93	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
94	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
95	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
96	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
97	C	Option C is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
98	A	Option A is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
99	D	Option D is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
100	B	Option B is correct. Oleum is formed by absorbing sulfur trioxide into 98% concentrated sulfuric acid: $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$. Its systematic IUPAC name is disulfuric(VI) acid.	[1]
101	C	Option C is correct. Molecular nitrogen has a triple covalent bond consisting of one strong sigma bond and two pi bonds with an enormous bond enthalpy of 945 kJ/mol. In addition, the homonuclear diatomic molecule is completely non-polar, offering no partial charges (δ^+ or δ^-) to attract attacking reagents. This results in an exceptionally high activation energy barrier for chemical reactions at room temperature.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
102	A	Option A is correct. Nitrogen dioxide acts as a homogeneous redox catalyst in the troposphere via two linked steps: Step 1: $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$. Step 2: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$. The regenerated NO_2 can then oxidise another molecule of SO_2 , rapidly driving the formation of SO_3 (which dissolves to form H_2SO_4 acid rain).	[1]
103	D	Option D is correct. Heating an ammonium salt with a base displaces ammonia: $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})$. Ammonia is an alkaline gas with a characteristic pungent smell that dissolves in the moisture of damp red litmus paper to produce OH^- ions, turning the paper blue.	[1]
104	B	Option B is correct. The direct combination of N_2 and O_2 to form NO is endothermic (+180 kJ/mol). Because breaking the $\text{N}\equiv\text{N}$ triple bond requires 945 kJ/mol of energy, the reaction rate at room temperature is effectively zero. Only inside car combustion chambers during the electrical spark (temperatures > 2000 °C) do colliding molecules have sufficient kinetic energy to overcome this immense activation energy barrier.	[1]
105	A	Option A is correct. Acidic SO_2 reacts with basic CaCO_3 to form calcium sulfite: $\text{CaCO}_3 + \text{SO}_2 \rightarrow \text{CaSO}_3 + \text{CO}_2$. Air is then bubbled through the slurry (forced oxidation) to convert CaSO_3 into hydrated calcium sulfate: $2\text{CaSO}_3 + \text{O}_2 + 4\text{H}_2\text{O} \rightarrow 2(\text{CaSO}_4 \cdot 2\text{H}_2\text{O})$. This synthetic gypsum is of high purity and is widely used to manufacture plasterboard for the building industry.	[1]
106	C	Option C is correct. The reaction $\text{SO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4$ is violently exothermic ($\Delta H = -130 \text{ kJ/mol}$). The heat generated instantly boils water into steam, producing a dense, highly acidic aerosol mist of microscopic droplets that escapes through exhaust vents. Absorbing SO_3 into 98% H_2SO_4 forms liquid oleum ($\text{H}_2\text{S}_2\text{O}_7$) smoothly and quietly, which is then diluted safely with water to produce concentrated sulfuric acid.	[1]
107	B	Option B is correct. Marble and limestone consist of calcium carbonate (CaCO_3). Atmospheric sulfuric acid in acid rain reacts according to: $\text{CaCO}_3(\text{s}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{CaSO}_4(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$. The calcium sulfate formed is porous and significantly more soluble than CaCO_3 , causing crumbling and irreversible erosion of historical sculptures.	[1]
108	D	Option D is correct. A three-way catalytic converter simultaneously promotes: (1) Reduction of nitrogen oxides to harmless nitrogen gas ($2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$), and (2) Oxidation of toxic carbon monoxide and unburnt hydrocarbons to carbon dioxide and water ($2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$; $\text{C}_x\text{H}_y + (x + y/4)\text{O}_2 \rightarrow x\text{CO}_2 + y/2\text{H}_2\text{O}$).	[1]
109	A	Option A is correct. Oleum (fuming sulfuric acid) is formed when SO_3 is dissolved in concentrated H_2SO_4 : $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$ (pyrosulfuric acid or disulfuric acid). Diluting oleum with water yields sulfuric acid: $\text{H}_2\text{S}_2\text{O}_7 + \text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4$.	[1]
110	C	Option C is correct. In NH_3 , the central nitrogen atom is surrounded by 3 bonding pairs and 1 non-bonding lone pair. The stronger lone pair – bonding pair repulsion compresses the H-N-H bond angles to $\sim 107.0^\circ$ (trigonal pyramidal). When the lone pair forms a dative bond with H^+ to produce NH_4^+ , nitrogen is surrounded by 4 equivalent bonding pairs and 0 lone pairs. All four bonding pairs repel equally, forming a regular tetrahedral ion with bond angles of exactly 109.5° .	[1]